

WEEK 4

Internet Programming and Web Development

HTML5 Advanced

Forms & Validation · Multimedia Elements · Accessibility & ARIA

☐ HTML5 Forms & Input Types

Forms, input types, labels, fieldsets

☐ Validation Attributes

required, pattern, min/max, minlength

☐ Multimedia Elements

audio, video, canvas

Accessibility & ARIA

WCAG, aria-*, roles, keyboard nav

LEARNING OBJECTIVES

By the end of this lesson, you can:

PI 1

Build complete HTML5 forms using appropriate input types for each field

PI 2

Apply validation attributes (required, pattern, min/max) without JavaScript

PI 3

Embed <audio> and <video> multimedia elements with fallback content

PI 4

Explain the purpose of <canvas> and when JavaScript is required to use it

PI 5

Add ARIA attributes to improve form accessibility for screen reader users

PI 6

Identify the 4 WCAG 2.1 principles and apply them to HTML structure

TOPIC A — HTML5 Form Structure

<form>, <fieldset>, <legend>, <label>, <button>

THE <form> ELEMENT

action — WHERE form data is sent (URL)

method — HOW it is sent:

GET → data visible in URL (searches)

POST → data hidden (logins, payments)

<label for='id'>

Links label text to its input field

Clicking label focuses the input

Screen readers announce it aloud

<fieldset> + <legend>

Groups related inputs visually

<legend> = caption for the group

Improves keyboard + screen reader UX

```
<form action='/submit' method='post'>

  <fieldset>
    <legend>Personal Info</legend>

    <label for='fname'>Full Name:</label>
    <input type='text' id='fname'
           name='fname'>

    <label for='email'>Email:</label>
    <input type='email' id='email'
           name='email'>

  </fieldset>

  <fieldset>
    <legend>Preferences</legend>
    <!-- checkboxes, radios here -->
  </fieldset>

  <button type='submit'>Send</button>
  <button type='reset'>Clear</button>

</form>
```

TOPIC A — HTML5 Input Types

Use the correct type — it activates free mobile keyboards, date pickers, color pickers, and more

`type="text"`

General single-line text

`type="url"`

URL format validation

`type="file"`

File upload button

`type="email"`

Email format validation

`type="date"`

Native calendar date picker

`type="search"`

Search field with clear ×

`type="password"`

Masked characters

`type="time"`

Time picker

`type="checkbox"`

Individual on/off toggle

`type="number"`

Numeric spinner + up/down

`type="color"`

Native color picker — returns hex

`type="radio"`

One choice from a group

`type="tel"`

Phone — phone keyboard on mobile

`type="range"`

Slider (min, max, step)

`type="hidden"`

Data sent with form, not shown

TOPIC B — HTML5 Validation Attributes

Server-quality form validation with zero JavaScript required

KEY INSIGHT: HTML5 validation runs in the browser BEFORE data is sent to the server — no JavaScript needed.

Attribute	Value	Purpose	Example Code
required	No value	Field must not be empty on submit	<code><input type='email' required></code>
placeholder	Text string	Greyed hint text inside field — disappears on typing	<code><input placeholder='Enter email'></code>
pattern	Regex string	Value must match the regular expression	<code><input pattern='[0-9]{11}'></code>
minlength	Number	Minimum character count (text inputs)	<code><input minlength='8'></code>
maxlength	Number	Maximum character count (text inputs)	<code><input maxlength='20'></code>
min	Number/date	Minimum value for number or date inputs	<code><input type='number' min='1'></code>
max	Number/date	Maximum value for number or date inputs	<code><input type='number' max='100'></code>
step	Number	Increment for number/range inputs	<code><input type='range' step='5'></code>

TOPIC A — More Form Elements

<select>, <textarea>, radio buttons, checkboxes — the essential form toolkit

```
<!-- DROPDOWN SELECT -->
<label for='subj'>Subject:</label>
<select id='subj' name='subj'>
  <option value=''>-- Choose --</option>
  <optgroup label='Technical'>
    <option value='bug'>Bug Report</option>
    <option value='feat'>Feature</option>
  </optgroup>
  <option value='other'>Other</option>
</select>
```

```
<!-- MULTI-LINE TEXT -->
<label for='msg'>Message:</label>
<textarea id='msg' name='msg'
  rows='5' cols='40'
  required placeholder='Your message...'
></textarea>
```

```
<!-- RADIO BUTTONS (grouped by name) -->
<fieldset>
  <legend>Gender</legend>
  <label>
    <input type='radio' name='gender'
      value='male'> Male
  </label>
  <label>
    <input type='radio' name='gender'
      value='female'> Female
  </label>
</fieldset>
```

```
<!-- CHECKBOX (independent) -->
<label>
  <input type='checkbox'
    name='subscribe' value='yes'>
  Subscribe to newsletter
</label>
```

KEY RULE — Radio Buttons: All radio buttons in the SAME GROUP must share the same name attribute. The value attribute tells the server WHICH option was chosen. Checkboxes are independent — each has its own name.

TOPIC C — Multimedia Elements

<audio>, <video>, <source>, fallback content, <canvas>

AUDIO ELEMENT

```
<audio controls>
  <source src='song.mp3'
    type='audio/mpeg'>
  <source src='song.ogg'
    type='audio/ogg'>
  Your browser does not
  support audio playback.
</audio>
```

controls	Shows play/pause/volume controls
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autoplay	Plays on load (+ muted required)
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loop	Restarts after ending
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muted	Starts with sound off
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VIDEO ELEMENT

```
<video controls width='400'
  poster='thumbnail.jpg'>
  <source src='clip.mp4'
    type='video/mp4'>
  <source src='clip.webm'
    type='video/webm'>
  Your browser does not
  support video playback.
</video>
```

controls	Browser play/pause/volume
-----------------	---------------------------

poster	Thumbnail image before play
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width / height	Pixel dimensions of player
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autoplay + muted	Required pair for autoplay
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loop	Repeat playback continuously
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<canvas id='myCanvas' width='400' height='200'> — Creates a blank drawing surface. JavaScript is REQUIRED to draw on it. Without JS: completely invisible box.

TOPIC D — Accessibility & ARIA

WCAG 2.1 Principles · ARIA Roles & Attributes · Keyboard Navigation

WCAG 2.1 — Four Principles (POUR)

Perceivable

All information must be presentable to users in ways they can perceive — alt text for images, captions for video

Operable

All UI must be navigable by keyboard alone — no mouse required. tabindex, focus order, skip links

Understandable

Content must be clear. Language declared (lang attribute). Error messages explain what went wrong

Robust

Content must work with current and future assistive technologies — valid HTML, ARIA, semantic structure

KEY ARIA ATTRIBUTES FOR FORMS

Attribute	Purpose	Example
<code>aria-required='true'</code>	Tells screen readers a field is mandatory — use WITH HTML required attribute	<code><input required aria-required='true'></code>
<code>aria-label='...'</code>	Provides accessible name when no visible label exists	<code><input type='search' aria-label='Search site'></code>
<code>aria-describedby='id'</code>	Points to an element that gives extra info about the field	<code><input aria-describedby='hint'> 11 digits</code>
<code>aria-hidden='true'</code>	Hides decorative elements from screen readers	<code>[icon]</code>
<code>aria-invalid='true'</code>	Announces a validation error to assistive tech	<code><input aria-invalid='true'></code> (set with JS after submit)
<code>role='...'</code>	Gives semantic meaning: role='main', role='navigation', role='alert'	<code><div role='alert'>Error: Email required</div></code>

LAB ACTIVITY — Accessible Contact Form

contact-form.html | 50 points | 50 minutes

R1 — 5 ptsHTML5 Boilerplate DOCTYPE, html lang, meta charset, meta viewport, title	R2 — 10 ptsForm + 9 Input Types text, email, tel, select, textarea, radio×3, checkbox, submit, reset	R3 — 8 ptsValidation Attributes required, minlength, pattern=[0-9]{11}, placeholder, min/max on age	R4 — 7 ptsLabels + Fieldsets Every input has <label for='id'>, 2 fieldsets with legends
R5 — 5 ptsAudio Element <audio controls> + <source> + fallback text	R6 — 5 ptsVideo Element <video controls poster width> + 2 <source> + fallback	R7 — 5 ptsARIA Attributes aria-required, aria-label, aria-describedby, aria-hidden	R8 — 5 ptsSemantic Structure <header>, <nav>, <main>, <footer>, <h1>, <h2>

WEEK 4 — SUMMARY

What you learned today



Forms & Inputs

`<form>`, `<input>` (15+ types), `<select>`, `<textarea>`, `<fieldset>/<legend>` — the complete form toolkit



Validation

required, placeholder, pattern, minlength/maxlength, min/max — browser enforces rules, no JS needed



Multimedia

`<audio>` and `<video>` with controls, source, fallback. `<canvas>` = blank JS drawing surface



Accessibility

WCAG POUR principles. ARIA: aria-required, aria-label, aria-describedby, aria-hidden

NEXT: Week 5 — CSS3 Fundamentals (Selectors, Box Model, Typography, Colors)